

Specialisation Tracks

Day-wise plan — high-level topic coverage

Four tracks · ten days each (two weeks) · specialisation only

Track 1 — Data Management

10-day specialisation · Week 1 (Days 1–5) + Week 2 (Days 6–10)

WEEK 1 — CORE

Day 1 — The Modern Data Foundation & the Agentic Data Value Chain

Major topics: The data value chain, end to end · The lakehouse in 2026 · Open table formats · Enterprise platforms · Data products & data contracts · AI-ready data · Storage & format choices · Catalogs as the control plane · Ingestion patterns · Modelling for analytics & AI · Cost & platform economics · The lakehouse for your lane

Day 2 — Agentic Data Engineering & Pipelines

Major topics: Orchestration & transformation backbone · The agentic data pipeline · Self-healing pipelines · NL → pipeline code · Multi-agent pipeline teams · Trust & graduated autonomy · Claude for data engineering · Wiring agents to the stack · Pipeline memory · Keep the model out of the data path · Guardrails for autonomy · Legacy compatibility

Day 3 — Data Quality, Governance, Catalog & MDM — Agentic

Major topics: Data quality & contracts · Data observability · The catalog as control plane · Governance for data and AI · Business semantics · Privacy & compliance · Master Data Management · Agentic MDM · Reference & active metadata · Quality automation · Stewardship workflows · Governance for your lane

Day 4 — Serving Data to AI & Agents — Semantic Layer, Text-to-SQL & Retrieval

Major topics: The semantic layer · Text-to-SQL / NL data access · Keep data safe · Permission-aware serving · Grounding agents on the lakehouse · Serving AI consumers · Agentic-RAG over data · GraphRAG over entities · Permission-aware retrieval · Citations, lineage & trust · Evaluation · Cost & latency

Day 5 — Productionisation, DataOps, Security, Cost & Capstone

Major topics: DataOps & CI/CD for data · Observability & SLOs · Reliability & scale · Security & access · Cost & FinOps · Responsible AI & regulation · End-to-end hardening · Audit & compliance packs · Runtime AI governance · Value & ROI · Capstone framing · Specialisation & next steps

WEEK 2 — ADVANCED & CAPSTONE

Day 6 — Real-Time & Streaming Data Products

Major topics: The 2026 streaming landscape · Streaming-first lakehouse · Shift-Left Architecture 2.0 · dbt meets Flink · Real-time for agentic AI · Freshness, cost & small files · Ingestion & processing · Stream → table · Event-driven agents · Freshness SLOs & observability · Reliability · Cost control

Day 7 — Advanced Agentic Data Orchestration & Autonomous Pipelines

Major topics: From self-healing to autonomous pipelines · Agent frameworks for data · NL → pipeline at scale · Pipeline memory & learning · Graduated autonomy & governance · A2A & cross-system agents · Orchestration graph · Tooling via MCP · Self-healing in depth · Observability for agents · Safety · Cost & scale

Day 8 — Advanced Governance, Active Metadata, Data Mesh & FinOps

Major topics: Active metadata & the catalog as control plane · Runtime AI governance · Data mesh & data products · Lineage & compliance at scale · FinOps for data & AI · Reliability & sovereignty · Contracts as code · Federated governance · Active-metadata automation · Audit & compliance packs · Cost governance · Sovereignty & residency

Day 9 — Data for AI at Scale: Feature Stores, Vector & Knowledge Graphs, Agent Memory

Major topics: Feature & embedding platforms · Agentic RAG & GraphRAG at scale · Semantic layer for AI · Agent memory infrastructure · Permission-aware serving · Real-time + AI · Ingestion → embed pipeline · GraphRAG build · Governed NL access · Caching & cost · Lineage & trust · Evaluation

Day 10 — Capstone Build & Shark Tank Finale

Major topics: Integration · Proof of outcomes · Demo engineering · ROI & business case · Governance & audit · Pitch craft · Pitch · Demo · Proof · Q&A · Scoring · Next steps

Track 2 — Customer Experience (CX)

10-day specialisation · Week 1 (Days 1–5) + Week 2 (Days 6–10)

WEEK 1 — CORE

Day 1 — Agentic CX Foundations & a Resolution Agent

Major topics: The CX agent landscape (2026) · Deflection vs resolution · The agentic CX loop · Knowledge & grounding · Human-in-the-loop & escalation · Trust & guardrails (intro) · Agent architecture · Resolution flows · Actions & confirmations · Persona & tone · Containment & escalation · Instrumentation

Day 2 — Conversation Design, Multi-Agent CX & Omnichannel

Major topics: Conversation design · Persona & policy · Multi-agent CX · Agent assist · Persistent memory · Omnichannel · Supervisor orchestration · Specialist agents · Agent-assist mode · Memory architecture · Channel adapters · QA hooks

Day 3 — Voice Agents — the Real-Time Stack

Major topics: The voice pipeline · STT / ASR · TTS · Orchestration · Turn-taking & interruptions · Build vs buy · Pipeline assembly · Latency engineering · Telephony · Reliability · Voice eval & QA · Compliance

Day 4 — Actions, Integration, Guardrails & Compliance

Major topics: Enterprise integration · Action design · Guardrails · Prompt-injection & untrusted content · Identity & permissions · Compliance · MCP integration · Transactional actions · Defence-in-depth · Escalation & safe failure · Policy-as-config · Compliance pack

Day 5 — Evaluation, ROI, Governance & Capstone

Major topics: CX evaluation · Continuous QA · Observability · ROI · Governance · From PoC to production · End-to-end hardening · Eval-gated rollout · Governance pack · ROI dashboard · Capstone framing · Specialisation & next steps

WEEK 2 — ADVANCED & CAPSTONE

Day 6 — Advanced Voice & Multimodal CX with Google Gemini

Major topics: From pipeline to native audio · Affective dialogue & proactive audio · Real-time multimodality · Production architecture · Tool use & grounding · Gemini vs the modular stack · Gemini Enterprise Agent Platform · Customer Engagement Suite · CX Agent Studio · Tooling & actions · Latency & reliability · Compliance

Day 7 — Proactive, Outbound & Multilingual CX Agents

Major topics: Proactive & outbound agents · Multilingual & localisation · Journey orchestration · Personalisation · Consent, compliance & brand safety · Measuring proactive value · Outbound orchestration · Memory across the journey · Localised personas · Safety rails · Hand-off · Analytics hooks

Day 8 — CX Analytics, Personalisation & Continuous Improvement

Major topics: Conversation analytics & insights · Continuous QA · Personalisation engines · Knowledge management · The improvement loop · Metrics that matter · Analytics pipeline · QA automation · Trace mining → goldens · Eval-gated updates · Personalisation in the loop · Dashboards

Day 9 — Enterprise CX at Scale: CCaaS, Reliability, Advanced Guardrails & Safety

Major topics: CCaaS integration · Reliability at scale · Advanced guardrails · Identity & permissions · Compliance at scale · Cost & capacity · Routing & handoff · Resilience engineering · Guardrail stack · Audit & governance · Capacity & cost · Security

Day 10 — Capstone Build & Shark Tank Finale

Major topics: Integration · Proof of outcomes · Demo engineering · ROI & business case · Governance & safety · Pitch craft · Pitch · Demo · Proof · Q&A · Scoring · Next steps

Track 3 — AI Engineering & Generative AI

10-day specialisation · Week 1 (Days 1–5) + Week 2 (Days 6–10)

WEEK 1 — CORE

Day 1 — GenAI Application Engineering Foundations

Major topics: The GenAI app stack · The 2026 model landscape · Working with model APIs · Prompt engineering for production · Structured outputs & tool calling · Cost & latency engineering · Service architecture · Prompt / version management · Output validation & repair · Caching & cost control · Secrets, config & deployment · Observability from day one

Day 2 — Retrieval, RAG & Context Engineering

Major topics: RAG architecture · Vector data · From RAG to Agentic RAG · GraphRAG & hybrid · Context engineering · Permission-aware retrieval · Ingestion & indexing pipeline · Retrieval quality · Reranking & fusion · Multimodal retrieval · Caching & cost · Grounding & citations

Day 3 — Model Adaptation — Prompting vs RAG vs Fine-Tuning

Major topics: The decision framework · Supervised fine-tuning & PEFT · Data curation · Tooling · Fine-tuning embeddings for RAG · Multimodal & MoE adaptation · Running a LoRA / QLoRA job · Evaluation vs baseline · Merge & export · Catastrophic forgetting & safety · When NOT to fine-tune · Packaging the decision

Day 4 — Inference, Serving, Performance & Cost

Major topics: Serving stacks · KV-cache & memory · Latency engineering · Cost optimisation · Self-host vs API · Cloud deployment · Throughput tuning · Routing & cascades · Reliability · Scaling · Cost governance · Deployment & rollout

Day 5 — Evaluation, Guardrails, LLMOps & Capstone

Major topics: Evaluation · Observability · Guardrails · Red-teaming · LLMOps · Governance · End-to-end hardening · Online evals & monitoring · Cost & reliability SLOs · Responsible AI · Capstone framing · Specialisation & next steps

WEEK 2 — ADVANCED & CAPSTONE

Day 6 — Advanced Agentic Architectures & Multi-Agent Systems

Major topics: The five-layer agent stack · Orchestration frameworks (2026) · Single vs multi-agent · Coordination patterns · Interoperability · Agent evaluation · Supervisor orchestration · Specialist design · State & handoffs · Tooling via MCP · Cost & latency · Trajectory evaluation

Day 7 — Context Engineering, Memory & Long-Horizon Agents

Major topics: Why context isn't memory · Memory architectures (2026) · Memory types · Context compression · Long-horizon reasoning · Open problems · Memory store design · Retrieval into context · Personalisation · Forgetting & staleness · Evaluation · Privacy & safety

Day 8 — Advanced Adaptation: Reasoning, Preference Tuning, Distillation & Small Models

Major topics: Beyond supervised fine-tuning · Reasoning fine-tuning · Distillation · Small specialised models · Model merging · Build vs buy · Preference / reasoning training · Distillation pipeline · Evaluation vs baseline · Serving the small model · Safety & regression · The decision record

Day 9 — Production LLMOps & AgentOps at Scale — Reliability, Observability & Cost

Major topics: Online evaluation · Observability at scale · Reliability engineering · Cost governance at scale · Security & continuous red-teaming · Incident response · Eval-gated CD · Trace mining · Reliability SLOs · Cost dashboards · Continuous security · Governance

Day 10 — Capstone Build & Shark Tank Finale

Major topics: Integration · Proof of outcomes · Demo engineering · ROI & business case · Governance & safety · Pitch craft · Pitch · Demo · Proof · Q&A · Scoring · Next steps

Track 4 — Low-Code / No-Code

10-day specialisation · Week 1 (Days 1–5) + Week 2 (Days 6–10)

WEEK 1 — CORE

Day 1 — Low-Code Agentic Foundations & a First Workflow

Major topics: The 2026 landscape · No-code vs low-code vs pro-code · Anatomy of a low-code agent · Deterministic vs agentic · Choosing a platform · Prompting & structured outputs in low-code · Platform setup · LLM nodes · Integrations · Branching & logic · Human-in-the-loop · Run logs & debugging

Day 2 — Agent Builders — RAG, Tools & Memory

Major topics: Agent builders compared · Built-in RAG · Tools & actions · Memory · Multi-agent in low-code · Templates & speed · Knowledge ingestion · Retrieval tuning · Tool wiring · Memory & personalisation · Testing in-platform · Publishing

Day 3 — Integration, Actions & MCP across Systems

Major topics: Integration breadth · MCP in low-code · Actions & transactions · Deterministic + agentic · Error handling & reliability · Identity & permissions · Multi-system flows · Approvals & audit · Idempotency & rollback · Scheduling & triggers · Governance hooks · Cost & quota

Day 4 — Conversational & CX Assistants in Low-Code

Major topics: Assistant patterns · Channels · Persona & guardrails · Grounding + actions · Escalation & handoff · Governance for conversational agents · Build · Channel deploy · Persona & policy · Human-in-the-loop & escalation · Testing · Publish & monitor

Day 5 — Governance, Scale, Lock-in & Capstone

Major topics: Governance · Observability · Cost & quotas · Security & data · Lock-in & portability · The no-code → pro-code path · End-to-end hardening · Audit & compliance · ROI & value · Scale plan · Capstone framing · Specialisation & next steps

WEEK 2 — ADVANCED & CAPSTONE

Day 6 — Building on Microsoft Power Platform — Copilot Studio, Code Apps & AI Coding Agents

Major topics: Copilot Studio (2026) · The Dataverse backbone · Power Apps Code Apps (GA 2026) · Code Apps + agents · AI coding agents for Power Platform · Real-time voice & connectors · Authoring discipline · Code App structure · Code App ↔ Copilot Studio · AI-assisted authoring · Deterministic guardrails · Channels & packaging

Day 7 — Multi-Agent Orchestration, Computer Use & Autonomous Agents

Major topics: Multi-agent in low-code · Computer Use in Copilot Studio (GA May 2026) · Computer Use models & governance · Guided autonomy · Cross-platform interoperability · When to use computer use · Supervisor & handoffs · Computer-use runbook · Safety for computer use · Scoped autonomy · Audit & replay · Cost & reliability

Day 8 — Enterprise Governance, Security & ALM for Citizen Development

Major topics: Microsoft Agent 365 · Defender Agent SPM · Entra & Purview controls · Environments & ALM · Unified inventory · Compliance & residency · Control-plane operations · ALM pipeline · Least privilege · Audit & evidence · Sprawl control · Cost governance

Day 9 — Observability, Evaluation, Scale & the Pro-Code Boundary

Major topics: Evaluation in low-code · Observability · Reliability & scale · Cost & capacity at scale · The pro-code boundary · Hybrid architecture · Eval-gated change · Reliability engineering · Cost control · Pro-code decision record · Hybrid integration · Handover

Day 10 — Capstone Build & Shark Tank Finale

Major topics: Integration · Proof of outcomes · Demo engineering · ROI & business case · Governance & safety · Pitch craft · Pitch · Demo · Proof · Q&A · Scoring · Next steps